EXPERIMENT 5

DATA VISUALIZATION WITH MATPLOTLIB

AIM:

To visualize and analyze the given dataset using various plots such as line charts, bar graphs, pie charts, and histograms with the help of the Matplotlib library, for better interpretation and understanding of trends and patterns.

PREREQUISITES & REQUIREMENTS:

1. Computer with Python Installed
2. A dataset for Practice
3. Jupytor Notebook
4. Knowledge on Pandas & Matplotlib Libraries

MATPLOTLIB

1. Matplotlib is a popular python library for Data Visualization.
2. It can be used to draw 2-D plots from data and arrays.
3. It supports almost every plot types such as line plots, scatter plots, bar charts, histograms etc.
4. Many other visualization libraries such as seaborn, python built-in visualizations are directly built with Matplotlib.
5. Matlotlib is inspired by the plotting functions of MatLab programming language.
6. In Matplotlib, creating plots can be done with functional based methods and OOP based methods.
7. For more useful information about matplotlib - https://matplotlib.org/2.0.2/gallery.html

Step 1: Import the required libraries & dataset

In [1]:
```
# Import the libraries
import pandas as pd
import matplotlib.pyplot as plt
```

In [4]:
```
# Load the dataset
_241UDS0092_df = pd.read_csv(r"C:\Users\hi\Downloads\monthly_sales_matplotlib_dataset.csv")
```

In [5]:
```
# Info of the Dataset
_241UDS0092_df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 4 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Month          12 non-null     str
 1   Sales          12 non-null     int64
 2   Profit         12 non-null     int64
 3   Products_Sold  12 non-null     int64
dtypes: int64(3), str(1)
memory usage: 516.0 bytes
```

In [6]:
```
_241UDS0092_df.head()
```

Out[6]:

	Month	Sales	Profit	Products_Sold
0	January	45000	8000	320
1	February	52000	9500	350
2	March	48000	8700	330
3	April	61000	12000	410
4	May	58000	11000	390

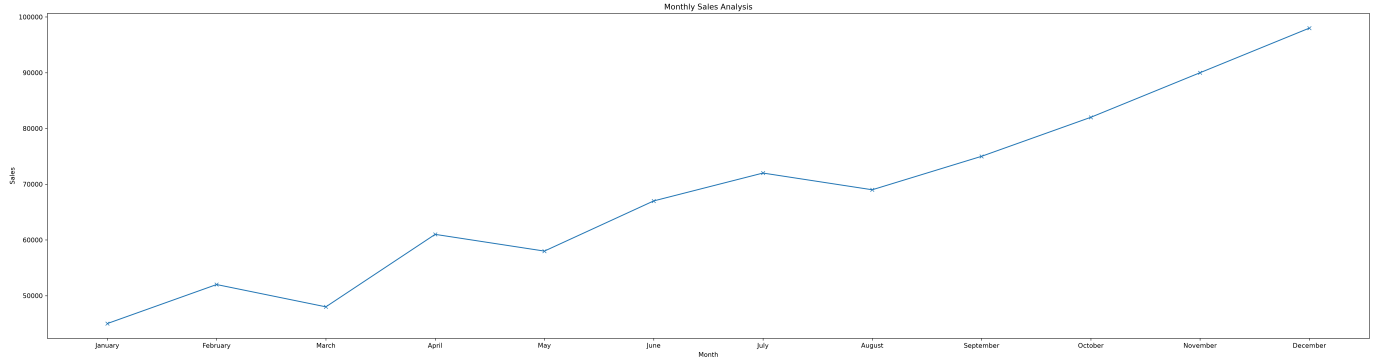
In [8]:
```
# Month wise data points in the dataframe
_241UDS0092_df.value_counts('Month')
```

Out[8]:
```
Month
January      1
February     1
March        1
April        1
May          1
June         1
July         1
August       1
September    1
October      1
November     1
December     1
Name: count, dtype: int64
```

Step 2: Visualize Data Using Various Plots

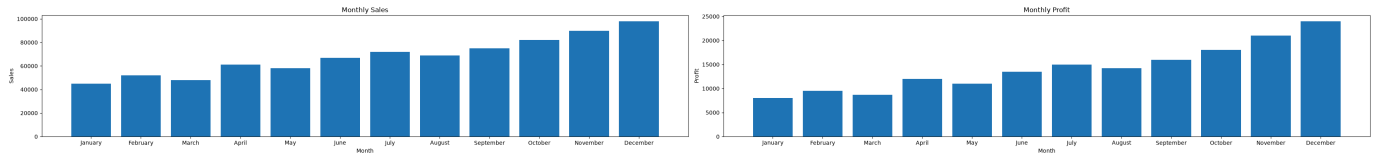
Line Plot — Monthly Sales

In [14]:
```
plt.figure(figsize=(30,8), dpi=250)
plt.plot(_241UDS0092_df['Month'], _241UDS0092_df['Sales'], marker='x')
plt.title('Monthly Sales Analysis')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.tight_layout()
plt.show()
```



Subplots — Compare Sales and Profit

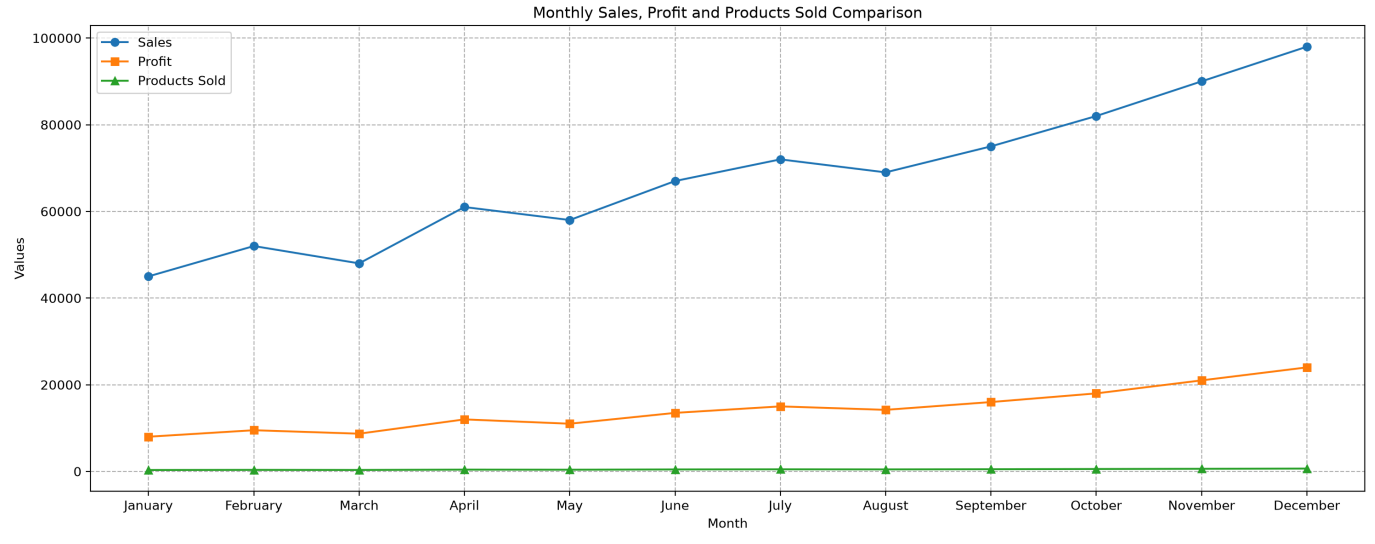
In [16]:
```
fig, axes = plt.subplots(1, 2, figsize=(35, 4))
axes[0].bar(_241UDS0092_df['Month'], _241UDS0092_df['Sales'])
axes[0].set_title('Monthly Sales')
axes[0].set_xlabel('Month')
axes[0].set_ylabel('Sales')
axes[1].bar(_241UDS0092_df['Month'], _241UDS0092_df['Profit'])
axes[1].set_title('Monthly Profit')
axes[1].set_xlabel('Month')
axes[1].set_ylabel('Profit')
plt.tight_layout()
plt.show()
```



Single Layout — Sales, Profit & Products Sold Comparison

In [13]:
```
plt.figure(figsize=(15,6), dpi=200)
plt.plot(_241UDS0092_df['Month'], _241UDS0092_df['Sales'], marker='o', label='Sales')
plt.plot(_241UDS0092_df['Month'], _241UDS0092_df['Profit'], marker='s', label='Profit')
plt.plot(_241UDS0092_df['Month'], _241UDS0092_df['Products_Sold'], marker='^', label='Product

plt.title('Monthly Sales, Profit and Products Sold Comparison')
plt.xlabel('Month')
plt.ylabel('Values')
plt.legend(loc='upper left')
plt.grid(True, linestyle='--')
plt.tight_layout()
plt.show()
```



RESULT:

Various data visualization techniques using the Matplotlib library were successfully implemented to analyze monthly sales, profit, and products sold using line charts, bar graphs, pie charts, and histograms.

In [ ]: